This assignment will not be graded and is only for practice.

Key points: 1

Definition of linear dependence: A set of vectors $v_1, v_2, ...v_n \in V$ is said to be linearly dependent if there exist scalars $a_1, a_2, ...a_n \in \mathbb{R}$, not all zero such that $a_1v_1 + a_2v_2 + ... + a_nv_n = 0$.

To check the vectors $v_1, v_2, ...v_n \in \mathbb{R}^m$ (with usual addition and scalar multiplication) are linearly dependent, we have to verify that the homogeneous system of linear equations $Ax = 0$ has infinitely many solutions, where the j^{th} column of A is v_j.

Definition of linear independence: A set of vectors $v_1, v_2, ...v_n \in V$ is said to be linearly independent if $a_1v_1 + a_2v_2 + ... + a_nv_n = 0$ implies that $a_i = 0$ for $i = 1, 2, ...n$.

To check the vectors $v_1, v_2, ...v_n \in \mathbb{R}^m$ (with usual addition and scalar multiplication) are linearly independent, we have to verify that the homogeneous system of linear equations $Ax = 0$ has only the trivial solution (i.e., $x = 0$), where the j^{th} column of A is v_j.

If S is a set containing zero vector, then the set S is a linearly dependent set.

Let S be a set of vectors from a vector space V. If an element of S is a scalar multiple of another element from the set S, then S is a linearly dependent set.

1) Vectors (2,1) and (6, 3) from $\mathbb{R}^2$ are **1 point**

- linearly independent.
- linearly dependent.
- None of the above.

No, the answer is incorrect.
Score: 0

Accepted Answers:

linearly dependent.

2) Suppose the set $\{(3,7,4), v\}$ is a linearly dependent set in the vector space $\mathbb{R}^3$ with usual addition and scalar multiplication. Which of the following vectors are possible candidates for v? **1 point**

- $(0,0,0)$
- $(5, \frac{35}{3}, \frac{20}{3})$
- $(1,0,0)$
- $(6,14,-8)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(0,0,0)$
$(5, \frac{35}{3}, \frac{20}{3})$

3) The set of vectors $\{(1,1,-1),(-1,1,1), \text{ and } (0,\frac{1}{2},0)\}$ in $\mathbb{R}^3$ is **1 point**

- linearly independent.
- linearly dependent.

- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

linearly dependent.

4) $M_{2\times 2}(\mathbb{R})$ denotes the vector space consisting of real square matrices of order 2 with usual matrix addition and scalar multiplication. Consider the following matrices: **1 point**

$$M_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, M_2 = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, M_3 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Which of the following options is(are) true?

[**Hint:** Form a system of linear equations using matrix equation $x\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + y\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ = 0]

- [ ] $\{M_1, M_2\}$ is a linearly independent set.
- [ ] $\{M_1, M_2, M_3\}$ is a linearly dependent set.
- [ ] $\{M_2, M_3\}$ is a linearly dependent set.
- [ ] $\{M_1, M_3\}$ is a linearly independent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{M_1, M_2\}$ is a linearly independent set.
$\{M_1, M_3\}$ is a linearly independent set.

Key points: 2

Given two vectors $(x_1, y_1),$ and $(x_2, y_2) \in \mathrm{R}^2$ construct $A = [\begin{matrix} x_1 & x_2 \\ y_1 & y_2 \end{matrix}]$. Then these vectors are linearly dependent if and only if $det(A) = 0$ and linearly independent if and only if $det(A) = 0$.

Given three vectors $(x_1, y_1, z_1), (x_2, y_2, z_2)$ and $(x_3, y_3, z_3) \in \mathrm{R}^3$ construct $A = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$.

Then these vectors are linearly dependent if and only if $det(A) = 0$ and linearly independent if and only if $det(A) = 0$

5) Vectors (2,3) and (4, 5) from R^2 are **_1 point_**

[**Hint:** Form the matrix using the vectors and calculate the determinant of the matrix.]

- ○ linearly independent.
- ○ linearly dependent.
- ○ None of the above.

No, the answer is incorrect.
Score: 0

Accepted Answers:

linearly independent.

6) Consider a set of vectors $S = \{(1,2), (-1,2), (3,1), (-3,1)\} \subset \mathrm{R}^2$. Choose the set of correct options. **_1 point_**

[**Hint:** Form the matrix using the vectors as the columns for each option and calculate the determinant of the matrix.]

- ☐ $\{(1,2), (-1,2)\}$ is a linearly dependent set.
- ☐ $\{(-1,2), (-3,1)\}$ is a linearly independent set.
- ☐ $\{(-3,1), (-1,2)\}$ is a linearly independent set.
- ☐ $\{(3,1), (-3,1)\}$ is a linearly dependent set.
- ☐ $\{(3,1), (1,2)\}$ is a linearly independent set.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\{(-1,2), (-3,1)\}$ is a linearly independent set.
$\{(-3,1), (-1,2)\}$ is a linearly independent set.
$\{(3,1), (1,2)\}$ is a linearly independent set.

7) The set of vectors $\{(1,1,0),(0,1,1),(1,0,1)\}$ is **1 point**

[**Hint:** Form the matrix using the vectors as the columns and calculate the determinant of the matrix.]

- linearly dependent.
- linearly independent.
- None of the above.

No, the answer is incorrect.
Score: 0

Accepted Answers:

linearly independent.

8) Suppose $\{(2,3,0),(0,1,0),v\}$ is a linearly dependent set in the vector space R^3 with usual addition and scalar multiplication. Which of the following vectors are possible candidates for v? **1 point**

- [ ] $(2,3,1)$
- [ ] $(1,2,0)$
- [ ] $(1,3,0)$
- [ ] $(0,1,1)$
- [ ] $(\pi,e,0)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(1,2,0)$
$(1,3,0)$
$(\pi,e,0)$

9) Consider three vectors $v_1 = (1, 0, 0), v_2 = (1, 1, 0), v_3 = (1, 1, 1)$ in the vector space $\mathbb{R}^3$, with usual addition and scalar multiplication. Which of the following option is true? ***1 point***

[**Hint:** Find out the vectors of the given sets in the options and form the matrices using the vectors as the columns and calculate the determinant of the matrices.]

- $\{v_1, v_2, v_3\}$ is a linearly dependent set.
- $\{v_1 + v_2, v_1 + v_3, v_3\}$ is a linearly dependent set.
- $\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.

Key points: 3

Any set of n vectors in $\mathbb{R}^2$ with $n \geq 3$ is linearly dependent.
Any set of n vectors in $\mathbb{R}^3$ with $n \geq 4$ is linearly dependent.
Any set of $n + 1$ vectors in $\mathbb{R}^n$ is linearly dependent.
Any set containing a linearly dependent set is linearly dependent.
If a set is linearly independent, then every subset of it, is linearly independent.

10) The subset $S = \{(3,1),(1,2),(-3,1)\}$ of R^2 is

1 point

[**Hint:** Any set of n vectors in R^2 with $n \geq 3$ is linearly dependent.]

- linearly dependent.
- linearly independent.
- None of the above.

No, the answer is incorrect.
Score: 0

Accepted Answers:

linearly dependent.

11) Set $S = \{(1,1,-1),(-1,1,1),(0,\frac{1}{2},0),(0,1,-2),(1,0,-2)\}$ subset of R^3 is

1 point

[**Hint:** Any set of n vectors in R^3 with $n \geq 4$ are linearly dependent.]

- linearly dependent.
- linearly independent.
- None of the above.

No, the answer is incorrect.
Score: 0

Accepted Answers:

linearly dependent.

12) Let S be a linearly dependent subset of R^3. Which of the following options are true?

1 point

- [ ] $S \cup \{(1,0,0)\}$ must be linearly dependent.
- [ ] $S \cup \{v\}$ must be linearly dependent for any $v \in \mathrm{R}^3$.

- [ ] $S \setminus \{v\}$ must be linearly dependent for any $v \in S$.
- [ ] There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S \cup \{(1,0,0)\}$ must be linearly dependent.
$S \cup \{v\}$ must be linearly dependent for any $v \in \mathbb{R}^3$.
There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

Your score is: 0/12